



Figure 4: Inspecting elements

examination from a large pool of numbers (usually roll numbers). CBSE is one of the boards for education in India and the result day breaks many hearts. Let's write a simple script to retrieve the exam's results list for a given range of numbers.

Disclaimer: The following scripts are presented for educational purposes only. While using them, use the resultant data with utmost care and responsibility, and respect the privacy of others. Neither the author nor OSFY are responsible for any misuse.

For the sake of simplicity, let's target the 2013 results of CBSE Class XII.

Here's the complete script:

```
require 'watir-webdriver'
range = (1..90) # Replace numbers as per use
slacko = Watir::Browser.new :firefox #Makes a firefox pet
slacko.goto "http://resultsarchives.nic.in/cbseresults/
cbseresults2013/class12/cbse122013.htm" #Loads the target
page
range.each do |number|
  slacko.text_field(:name => 'regno').set number #Sets the
input with the current number in range
  slacko.button(:name => "B1").click #Clicks the submit button
  slacko.screenshot.save "#{number}.png" #Saves the screenshot
of result as number.png
  slacko.back #Loads the previous page
end
```

If there are no errors and the script runs fine, you will see a list of screenshots of results arranged by roll numbers. Since you will have a lot of data, use it carefully and don't use it to harm anybody. This could be particularly useful for schools that want to digitally archive the results of their students.

Going headless

The results retrieval script is good but browsers use resources.

We can replace Firefox or Chrome with a headless browser (one with no graphical interface) like *phantomjs* so as to save memory. Headless browsers like *phantomjs* can accomplish the same task with less resource consumption and also support screenshots.

Installing *phantomjs* is as easy as downloading the binary and placing it in the path.

The following commands will come handy:

#For 64 bit Linux systems

```
wget https://bitbucket.org/ariya/
phantomjs/downloads/phantomjs-1.9.8-
linux-x86_64.tar.bz2tar xvjf phantomjs-1.9.8-linux-x86_64
cd phantomjs-1.9.8-linux-x86_64/bin/
sudo ln -sf phantomjs /usr/local/share/phantomjs
sudo ln -sf phantomjs /usr/local/bin/phantomjs
sudo ln -sf phantomjs /usr/bin/phantomjs
```

If all goes right, a new *phantomjs* instance can be created by the following code:

```
require 'watir-webdriver'
slacko = Watir::Browser.new :phantomjs
#Do crazy things with slacko :) 
```

References

- [1] Ruby Installation Documentation: <https://www.ruby-lang.org/en/documentation/installation/>
- [2] Ruby Installer and Pk for Windows (Pk is no longer maintained, so proceed with care): <http://misheska.com/blog/2013/03/10/using-pk-to-manage-multiple-versions-of-ruby-on-windows/>
- [3] Uru Wiki: <https://bitbucket.org/ionforums/uru/wiki/Home>
- [4] Rbenv on Mac and Linux: <http://misheska.com/blog/2013/06/15/using-rbenv-to-manage-multiple-versions-of-ruby/>
- [5] Link to download the Chrome driver: <http://chromedriver.storage.googleapis.com/index.html>
- [6] 1 Watir cheat sheets and references: <https://awetest.zendesk.com/hc/en-us/articles/201883796-Watir-Webdriver-CheatSheet>
- [6] 2 <https://github.com/watir/watir/wiki/Cheat-Sheet>
- [7] Phantomjs installation instructions: <https://gist.github.com/julionc/7476620>
- [8] CBSE link used for demonstration: <http://resultsarchives.nic.in/cbseresults/cbseresults2013/class12/cbse122013.htm>
- [9] Phantomjs: <http://phantomjs.org/download.html>

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